

Clinical ECG: Atrial / Ventricular Arrhythmias

Ermin Tale DO, Internal Medicine PGY-2
Jacobi Medical Center/Albert Einstein College of Medicine
ermin.tale01@gmail.com

Todd Cohen MD, Chief of Cardiology and Director of Medical Device
Innovation
tcohen03@nyit.edu

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

1. Distinguish an arrhythmia from artifact.
2. List the causes of arrhythmias.
3. Distinguish a normal sinus rhythm from different bradyarrhythmias or tachyarrhythmias.
4. Sketch the algorithm to identify different tachyarrhythmias, identifying and creating a treatment plan along the way.

First Aid (2025): 293, 294,
315, 316

Rate & Rhythm: *What can go wrong?*

Arrhythmia= lack of normal rhythm

Dysrhythmia= “doesn’t work properly” rhythm

Tachyarrhythmia= fast arrhythmia

Bradyarrhythmia= slow arrhythmia



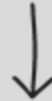
Why do we care about arrhythmias?

$$CO = SV \times HR$$

Brady



Tachy



Artifact

- Artifact occurs when something causes a disruption in monitoring.
- Some common causes are:
 - AC interference - causes 60 Hz artifact
 - Muscle contractions & tremors
 - Respiratory artifact- patient movement - wandering baseline
 - Loose electrode
 - Broken lead wire



Objective 1 Questions

Artifact can hide arrhythmia:

True

False

Objective 1 Questions

An 82 yo Male patient with a PMHx of type II diabetes, congestive heart failure, and generalized anxiety disorder is experiencing dyspnea and peripheral edema up to his knees. What is mechanism of these findings?

↑ CO → ↓ HR

↑ CO → ↓ TPR

↑ CO → ↓ SV

↓ CO → ↓ HR

↓ CO → ↓ TPR

↓ CO → ↓ SV

3 Mechanisms of Arrhythmias:

1. Automaticity

2. Triggered Activity

3. Re-entry

Objective 2 Questions

AV Nodal Reentry tachycardia's (AVNRT) mechanism of arrhythmia can be summarized with:

- a. an impulse travels back and re-excites the site of origin (reentry)
- b. generation of spontaneous electrical action potentials (automaticity)
- c. an underlying provocation (triggered activity)

Objective 2 Questions

This patient shows a 12 lead ECG with normal calibration at a sinus rate of 80 bpm. The axis is displaced to the left, with normal QRS and QT intervals. I don't see any significant ST elevations or T wave inversions." What major ECG finding is missing upon this report?

- a. A portion of orientation
- b. An interval
- c. Rhythm
- d. More than one of these answers

Bradyarrhythmias

- HR < 60 bpm
- Symptoms:
 - Dizziness, Lightheadedness
 - Palpitations, Pre-syncope
 - Syncope



Sinus Bradycardia

(SB)



STITUTE
GY
athic

- **Rate:** <60 bpm
- **Rhythm:** sinus (normal p wave precedes every QRS complex)
- **Intervals:**
 - **QRS:** Normal (.06-.12)
 - **Conduction:** P-R normal or slightly prolonged at slower rates
- **Signs/Sx:** Asymptomatic, lightheaded, dizziness, ↓BP, SOB, fatigue, syncope

Sinus Bradycardia (cont.)

- **Causes:**

- Common in athletes and during sleep
- Physiological response after vagal maneuver
- Intrinsic sinus node disease due to aging (sick sinus syndrome)
- Can result from medications (beta-blockers, etc)

- **Treatment**

- Usually self-limited → Treat the underlying cause
- **Atropine**, if the vitals signs are unstable
- **Pacemaker** for recurrent SB when associated w/ severe symptoms

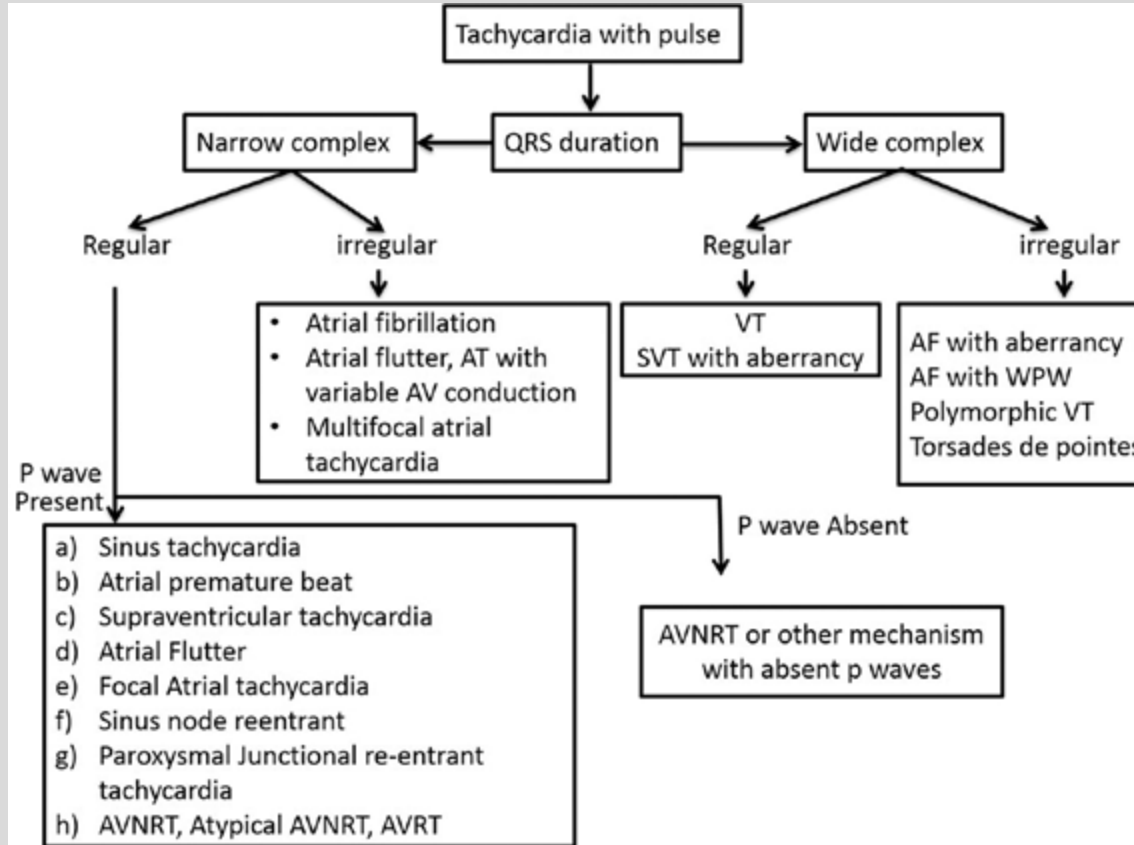


Tachyarrhythmias

- HR > 100 bpm
- Symptoms:
 - Asymptomatic, Palpitations
 - Shortness of breath, Fatigue
 - Syncope, Lightheadedness



Algorithm for Arrhythmias!



Arrhythmia originating above the ventricles (atria/nodes) = **narrow QRS**

Arrhythmia originating in the ventricles = **wide QRS**

Tachyarrhythmias

Narrow QRS

Regular

Supraventricular Tachycardia

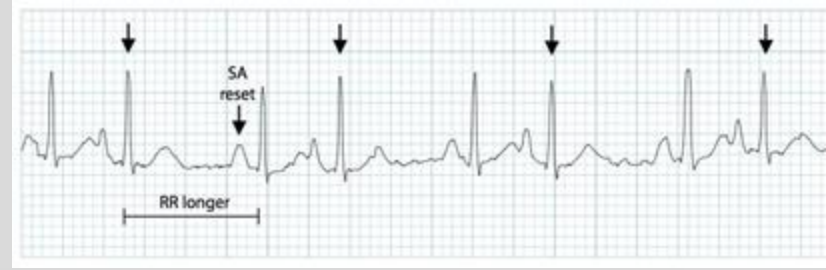
- A tachycardia that originates in or depends on parts above the ventricles
 - “supra” = above or earlier
 - Any tachycardia that is not from the ventricles
- Broad description of a rhythm – there are many more precise ways to describe a rhythm (such as some of the rhythms to follow)

Sinus Tachycardia

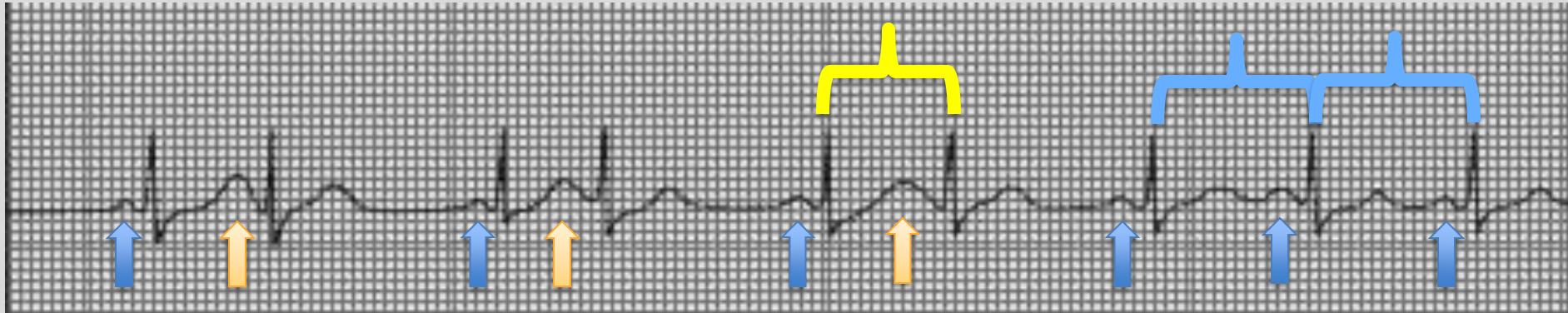


- **Rate:** > 100 bpm
- **Rhythm:** sinus tachycardia (normal p waves precede every QRS complex)
- **Intervals:**
 - **QRS:** normal
 - **Conduction:** P-R normal
- **Causes:** normal, physiologic response to pain, exercise, stress, infection, anemia, hyperthyroidism, dehydration or pulmonary embolism
- **Treatment:** includes identification of the underlying cause and correction.

Premature Atrial Contractions (PACs)



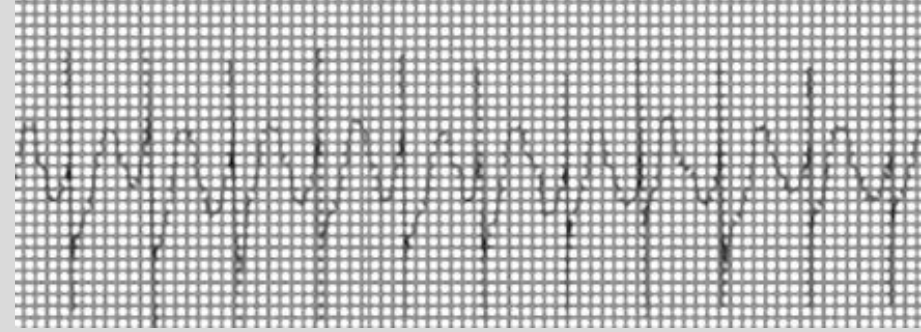
- **Premature activation of the atria from an atrial site other than the sinus node**
- **Rate:** normal or accelerated
- **Rhythm:** *irregular*
 - PAC P-waves usually have a different morphology than sinus P-waves because they originate from an ectopic pacemaker. Some can be embedded in T waves (or other places)
- **Intervals:**
 - P-R: ectopic beats may have a different P-R interval
 - QRS: usually normal
- **Causes:** occasional PACs are a normal occurrence; Occur more frequently in CHF, myocardial ischemia, COPD



↑ Normal p-waves

↑ Premature Atrial Contractions

Focal Atrial Tachycardia (FAT)



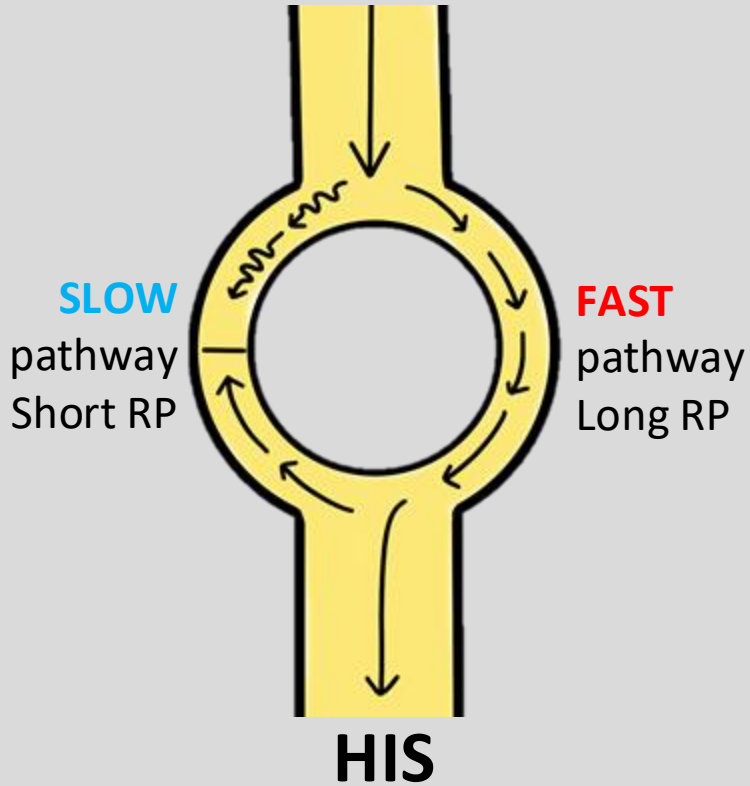
- Regular atrial rhythm originating outside the SA node with rate >100 bpm
 - **Specific atrial focus** is firing – not SA node
 - Most common location is the left atrium (63%)
- **Rate:** 150-250 bpm
- **Rhythm:** regular (p-waves almost hidden in T waves)
- **Intervals:** usually narrow QRS, normal PR
- Adenosine can reveal underlying atrial activity.

AV Nodal Reentrant Tachycardia (AVNRT)*

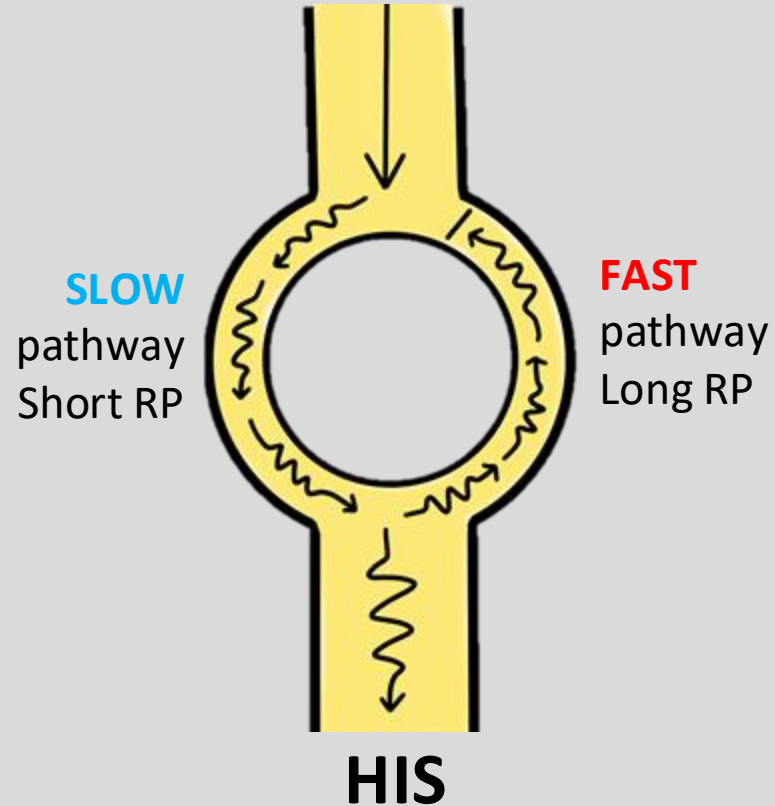
- Most common type of paroxysmal SVT
- Results from a **reentrant circuit** at the **site of the AV node** → depolarizes the atrium & ventricle simultaneously
- **Rate:** 150-250 bpm
- **Rhythm:** Regular. P-waves may be retrograde or buried
- The mechanism involves the AV node, so it must be disabled in order to “break” the rhythm (bring back into sinus rhythm)
 - Vagal maneuvers (e.g. carotid sinus massage, coughing, Valsalva)
 - Adenosine
- **Long-term treatment of choice is ablation.**



AVNRT



PAC





AVNRT (cont.)



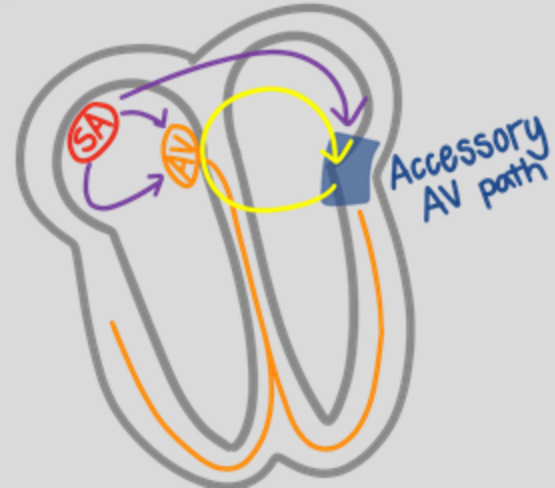
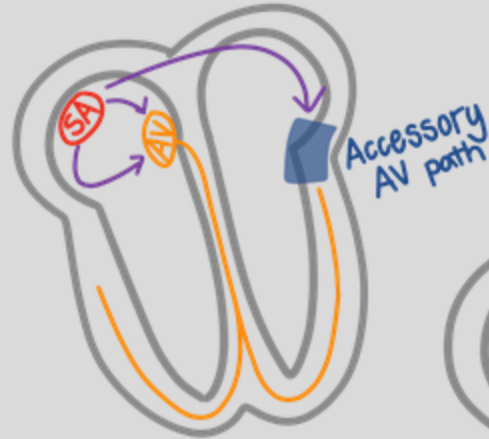
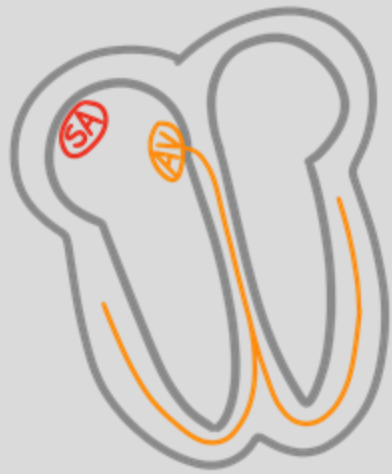
AV Reentrant Tachycardia (AVRT)*

- This involves a reentrant circuit as well:
 - **Accessory pathway:** abnormal connection between the atria and ventricles.
 - Two conduction pathways: the AV node (normal) and accessory pathway
- Retrograde P wave seen after the QRS
 - The accessory pathway can repolarize the p-wave
 - QRS can be either normal (orthodromic) or prolonged (antidromic) – depending on the pathways
- Treatment is similar to AVNRT



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

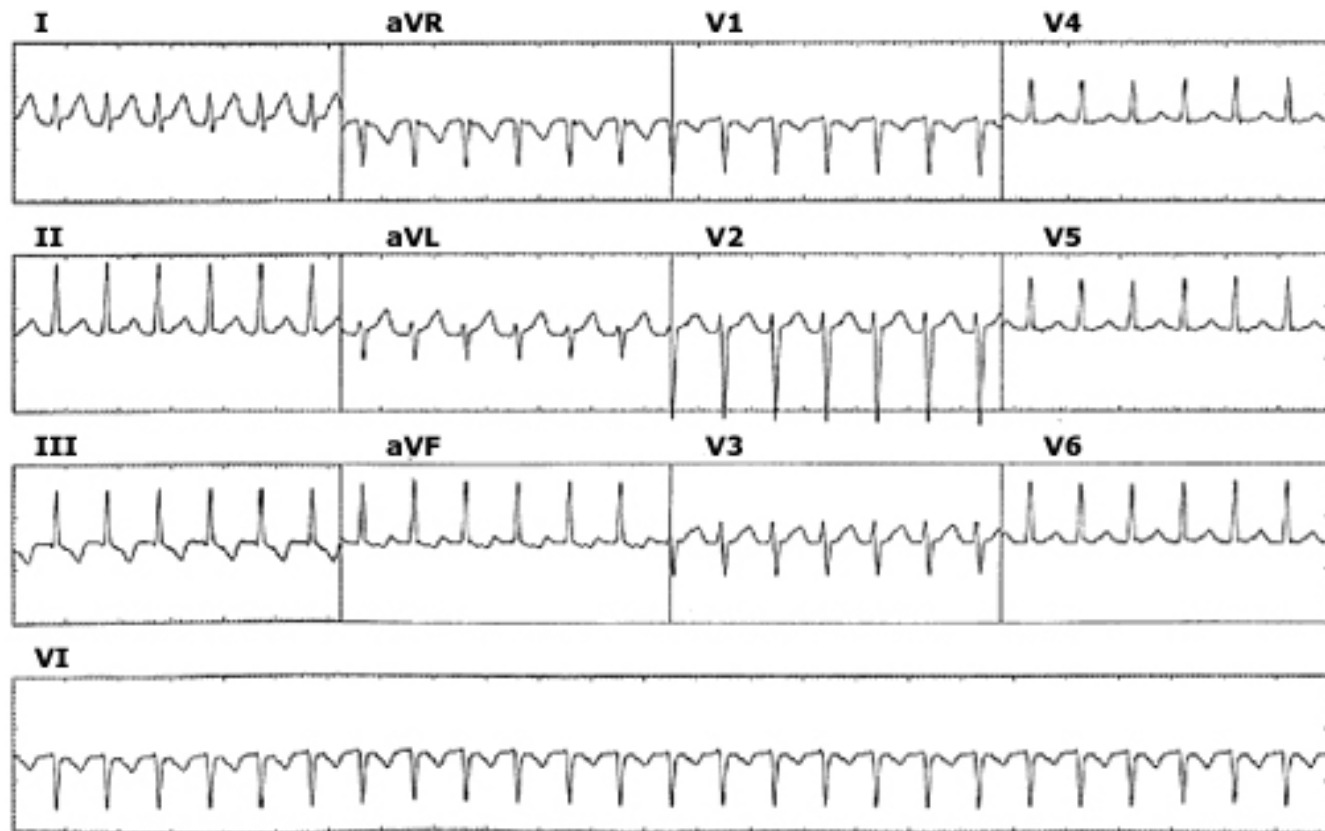


AVRT

AVRT

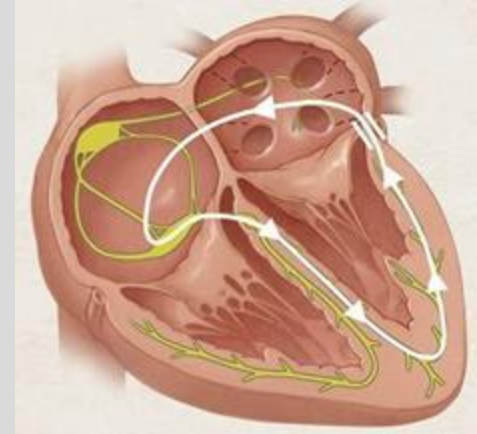
NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Wolff-Parkinson-White Syndrome**

- A form of AVRT
- Pre-excitation refers to early activation of the ventricles due to impulses bypassing the AV node via an accessory pathway.
- In WPW the accessory pathway is often referred to as the **Bundle of Kent**, or AV bypass tract, located on the **right ventricular wall**.



WPW Syndrome: ECG features & Treatment

- **ECG Findings**

- Shortened PR interval
- **Delta wave** – slurring slow rise of initial portion of the QRS
- QRS > 120ms (when pre-excited)
- ST Segment and T wave discordant changes – i.e. in the opposite direction to the major component of the QRS complex

- **Treatment:**

- Catheter ablation
- Vagal Maneuvers
- Meds: Procainamide

**** AVOID AV-NODAL SUPPRESSING AGENTS****

WPW – Delta Waves

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Tachyarrhythmias

Narrow QRS

Irregular

Atrial Fibrillation (A fib)

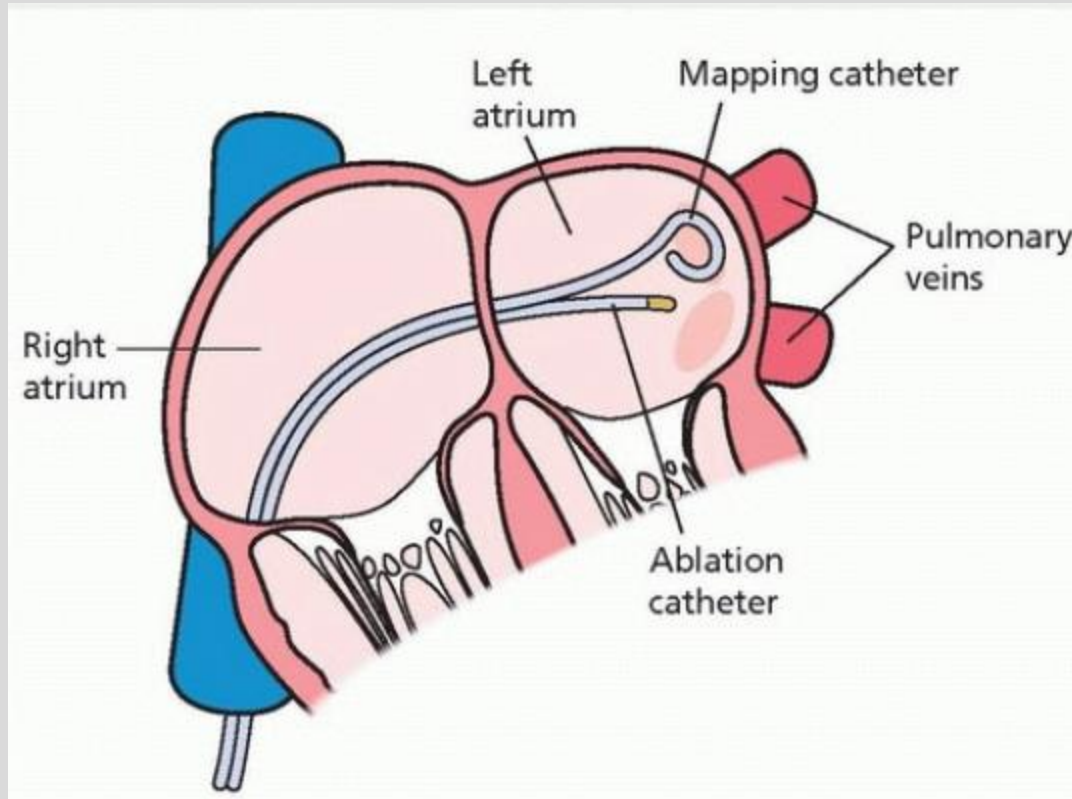
- Firing of multiple impulses from numerous ectopic pacemakers in the atria→ chaotic asynchronous activity in atrial tissue
 - Common location of focal origin is in LA near site of pulmonary veins
- **Rate:**
 - Atria is firing ~300-650 bpm – heart is quivering
- **Rhythm: irregularly irregular** (**NO** p-waves)
- **Intervals:** PR unmeasurable; usually narrow QRS
- **Causes:** a/w HTN (most common), COPD, CHF, underlying heart disease, aging, stress, infection, alcohol, thyrotoxicosis, caffeine



Management Options of A-Fib

- **Rate control = Rhythm control**
- **Rate control** = Beta-blockers (Metoprolol), Calcium Channel Blockers (Diltiazem), Digoxin
- **Rhythm control** = Amiodarone, Sotalol, Flecainide
- **Anticoagulation** prevent cerebrovascular accident = Warfarin, Enoxaparin, Rivaroxaban, Apixab, or Dabigatran
- Emergency = Electrical Cardioversion (DCCV)
- “Should do the trick”: Ablation → Pulmonary vein isolation

Catheter Ablation for A Fib



CHA2DS2-VASc Risk Stratification

Letter	Risk factor	Score
C	Congestive heart failure/LV dysfunction	1
H	Hypertension	1
A ₂	Age ≥ 75	2
D	Diabetes mellitus	1
S ₂	Stroke/TIA/thrombo-embolism	2
V	Vascular disease*	1
A	Age 65–74	1
S	Sex category (i.e., female sex)	1
	Maximum score	9

Congestive heart failure/LV dysfunction means LV ejection fraction $\leq 40\%$. Hypertension includes the patients with current antihypertensive medication. *Prior myocardial infarction, peripheral artery disease, aortic plaque. LV: left ventricular, TIA: transient ischemic attack

CHADS2-VASc Score:

0 $\rightarrow$ “low risk”

1 $\rightarrow$ “low-mod risk” $\rightarrow$ antiplatelet or

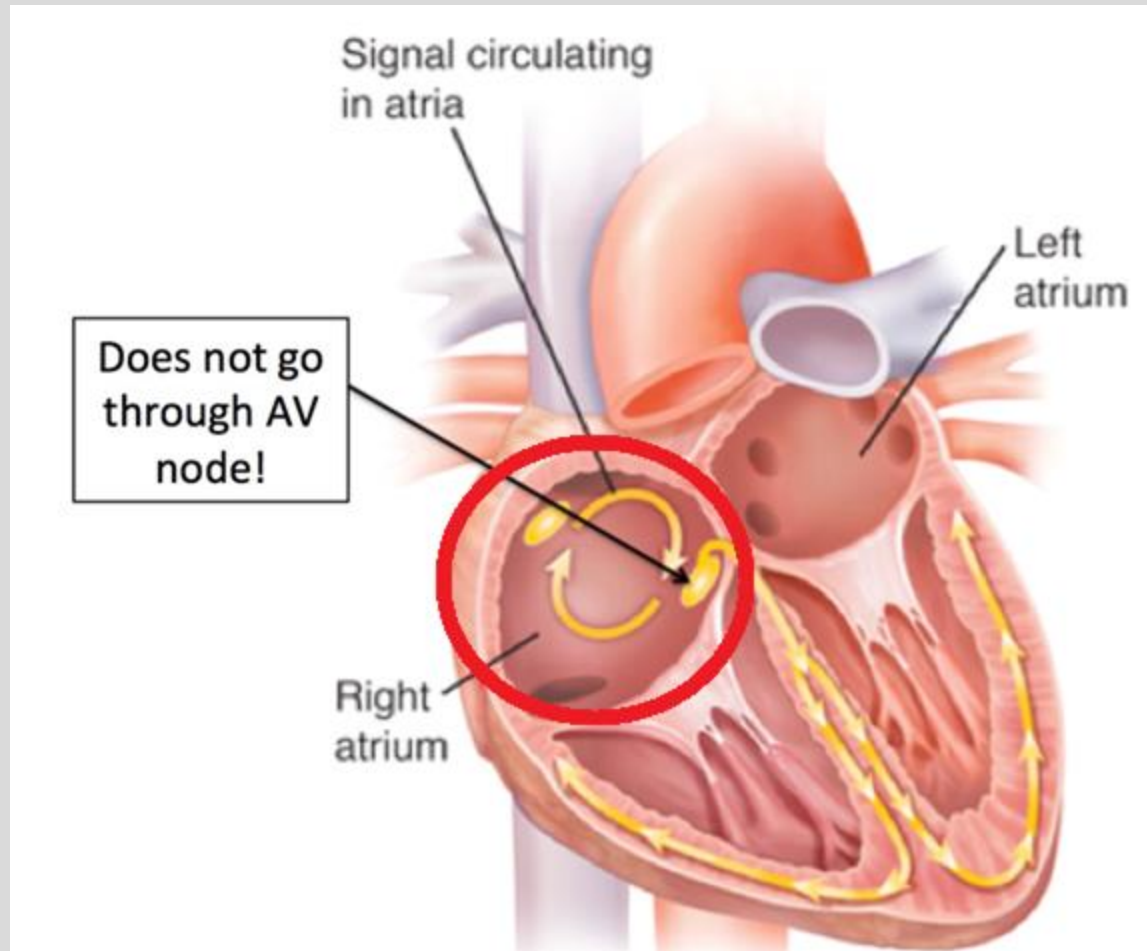
anticoagulant

$\geq 2 \rightarrow$ “mod-high risk” $\rightarrow$ anticoagulant



Atrial Flutter

- Atrial flutter is a type of SVT caused by a re-entry circuit within the right atrium that transverses the **cavo-tricuspid isthmus**
 - The length of the re-entry circuit corresponds to the size of the **right atrium**, resulting in a predictable atrial rate around 300 bpm (range 200-350)
- Ventricular rate is determined by the AV conduction ratio (degree of AV block)
- The most common AV ratio is 2:1, resulting in  ventricular rate of ~150 bpm



Atrial Flutter – ECG



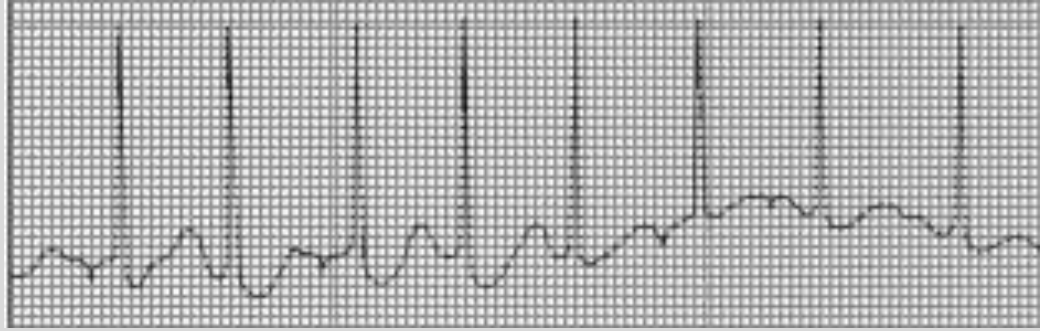
- Narrow complex tachycardia
- **Rate:** regular atrial activity at ~300 bpm, variable AV block
- **Rhythm:** Flutter waves (“**saw-tooth**” pattern) - II, III, aVF
 - Loss of the isoelectric baseline
 - Fixed or variable AV Block – **regular or irregular**
- **Intervals:** PR unmeasurable; usually a narrow QRS

Atrial Flutter Management

- **Difficult to control rate with pharmacology**
- **Anticoagulation**
- Cardioversion if unstable
- Ablation of the Cavo-tricuspid isthmus
 - Lower right atrium between IVC and tricuspid valve
 - **most common site*



Multifocal Atrial Tachycardia (MAT)



- Numerous atrial foci fire intermittently
- Can resemble A Fib/Flutter
- **Rate:** 100-250 bpm
- **Rhythm:** irregular (≥ 3 different p-wave shapes)
- **Intervals:** QRS narrow, PR interval variable
- **Cause:** most common in COPD/hypoxemia, hypomagnesemia and hypokalemia
- **Treatment:** Verapamil. Treat underlying cause

Multifocal Atrial Tachycardia (cont.)



Tachyarrhythmias

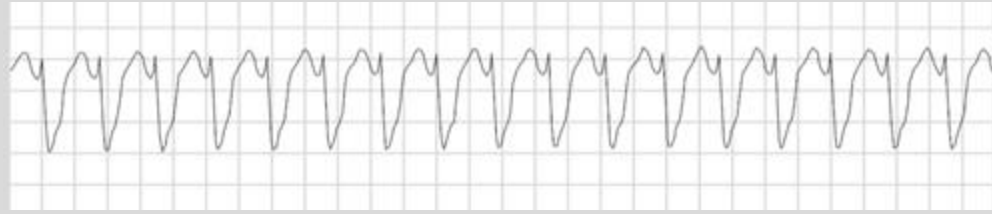
Wide QRS

Premature Ventricular Contractions (PVCs)



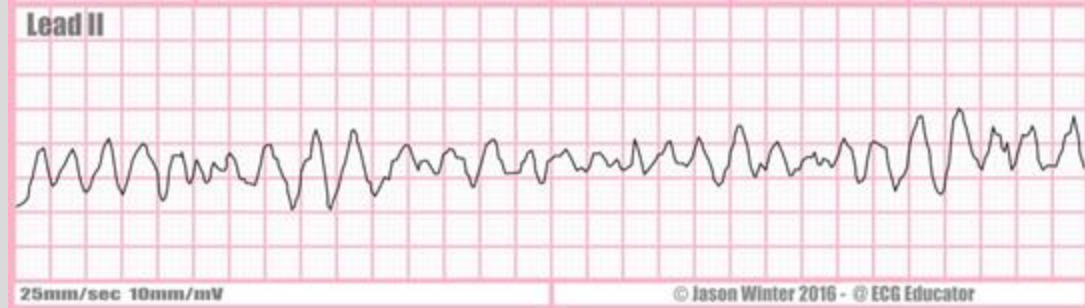
- Ectopic beats that originate below the branching point of the Bundle of His
- Impulse is generated below AV node → wide QRS usually without preceding P-wave
- **Causes:** a/w electrolyte abnormalities, hypoxia, hyperthyroidism, catecholamines, drug toxicities; can occur in both structurally normal or abnormal heart
- **Treat if:**
 - A/w acute MI
 - Occur as couplets, bigeminy, trigeminy
 - Are multifocal, or
 - Are frequent ($>6/\text{min}$) or symptomatic

Monomorphic Ventricular Tachycardia



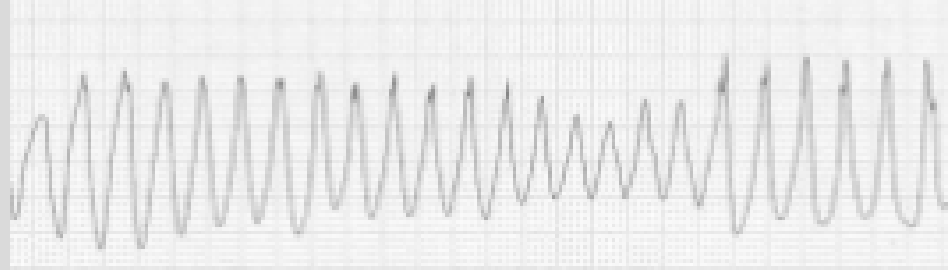
- Usually occurs in diseased hearts
- *Without treatment - deteriorates into ventricular fibrillation*
- **Rhythm:** p-waves are usually obscured/not present, are unrelated to the QRS complex
- **Causes:** CAD, acute MI, digitalis toxicity, CHF, ventricular aneurysms, cardiomyopathy, electrolyte abnormalities
- **Treatment:**
 - Symptomatic/Unstable: electrical cardioversion
 - Pharm: Amiodarone, Lidocaine

Ventricular Fibrillation (V fib)



- Chaotic electrical activity from the ventricles that causes the absence of cardiac output
- Treatment:
 - Immediate defibrillation and ACLS protocols
 - Identification and treatment of the underlying cause (often ischemic e.g. acute MI)

Torsades de Pointes



- Upward & downward deflection of QRS complexes around the baseline
- P-wave usually obscured if present, QRS complexes are wide, bizarre morphology
- **Causes:** drugs that prolong QT interval, MI, electrolyte imbalances (esp. hypo-Mg, hypo-K, hypo-Ca)
- Can progress to V Fib
- **Treatment:**
 - Synchronized cardioversion if patient unstable
 - IV Magnesium, IV Potassium
 - Overdrive pacing

HR 90 PVC 9 RESP 22 T1 38.3



Objective 3 & 4 Questions

A patient is unstable in ventricular tachycardia, with a pulse. You activate the appropriate hospital code. What needs to get done, as you are in the room with 3 nurses?

- a. Activate ACLS protocol
- b. Start chest compressions
- c. Prepare/ place the pads of a defibrillator onto the patient
- d. Give Amiodarone
- e. all of the above

Objective 3 & 4 Questions

A 69 year old female presents to the ER with palpitations. On EKG, you see sawtooth pattern, something she recurrently has every so often. Her cardiologist plans for gold standard treatment, terminating the possibility of this occurring in the future. What is this treatment?

- a. Ablation of myocytes near the pulmonary veins
- b. Ablation of myocytes near the cavo-tricuspid isthmus
- c. Long term propranolol
- d. Long term amiodarone
- e. Ablation of myocytes connecting the Bundle of Kent to the AV node

Summary Slide

- Premature Atrial Contractions (PAC's)- activation of the atria from an atrial **site other than the SA node**
- Focal Atrial Tachycardia- **Single atrial myocyte** (not the SA node) firing at a rate > 100 bpm
- AV nodal reentrant tachycardia (AVNRT)- **reentrant circuit at the AV node**
- AV reentrant tachycardia w/ accessory pathway (AVRT)- accessory pathway leads to two conduction pathways
- WPW- activation of the ventricles due to bypassing of the AV node via an accessory pathway- ie. **Bundle of Kent in the RV wall (ECG delta wave)**
- Atrial flutter- re-entry circuit within the right atrium transversing the **cavo-tricuspid isthmus** (ECG “**saw-tooth**” pattern)
- Atrial Fibrillation- atria are firing at 300-650 bpm originating from the **pulmonary veins (no p-waves)**
- Multifocal Atrial Tachycardia- multiple atrial myocytes firing creating **three distinct p-waves**
- Premature ventricular contraction (PVC's)- impulse generated **below the bifurcation of the His bundle**
- Monomorphic Ventricular Tachycardia- **uniform wide QRS complex**
- Ventricular Fibrillation- minimal cardiac output, requires **immediate defibrillation**
- Torsades de Pointes- associated w/ prolonged QT interval, MI, and electrolyte imbalances (first line txt- **IV Magnesium**)

How I would approach the lecture:

1. Distinguish an arrhythmia from artifact.
 - a. What is an arrhythmia?
 - b. What is artifact and what causes artifact?
 - c. Why do we care about tachy and bradyarrhythmias?
 - i. *Can alter CO₂ or oxygen demand!*

How I would approach the lecture:

2. List the causes of arrhythmias.
 - a. Review an approach to a rhythm strip
 - i. *Need to know how to identify abnormalities*
 - b. 3 Mechanisms of Arrhythmias
 - i. *Re-entry= AVNRT or AVRT*
 - ii. *Automaticity= meds alter rate*
 - iii. *Triggered Activity= Torsades de Pointes, V-fib, V-tach etc. (emergencies)*

How I would approach the lecture:

3. Distinguish a normal sinus rhythm from different bradyarrhythmias or tachyarrhythmias.

a. Discuss Sx, Causes & Tx of:

i. Sinus Bradycardia, Sinus Tachycardia,
“SVT”, PAC, FAT, AVNRT, WPW, A Fib, A
Flutter, MAT, PVC, V tach, V fib, Torsade

How I would approach the lecture:

4. Sketch the algorithm to identify different tachyarrhythmias, identifying and creating a treatment plan along the way.

- a. While we did the pathologies, we grouped them: walking through ID-ing:
 - i. Narrow vs Wide QRS
 - ii. Regular vs Irregular

Final Slide:

Questions:

Ermin Tale: ermin.tale01@gmail.com

Justin Edward: jedward@nyit.edu

Grace Robinson: grobin03@nyit.edu

More exposure → better EKG skills

-ECG Lectures **Mon 12-1 PM EST** with Dr. Cohen via Zoom -Will be sent out to the class each week by Dr. Cohen's Research Team

Feedback on the lecture:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>